

Maersk Landside Transportation Pricing Analysis

Project Summary

This project simulates a landside transportation pricing dataset and analyzes how pricing decisions affect win rate, customer profitability, CM1, and EBIT. The project is designed to reflect the analytical work expected in a Landside Transportation Product Analyst role, especially around pricing strategy, loss-making shipments, and balancing competitiveness with margin protection.

The summary follows common project summary elements such as project overview, objectives, results, and lessons learned, which are standard components of concise stakeholder reporting templates.

Business Goal

The business goal is to answer one core question: how should a landside transportation team price shipments so it can win enough business without damaging profitability. This matters because pricing above the market reduces win rate, while pricing too far below the market may protect volume but weaken CM1 and EBIT.

This project focuses on the same kind of decisions highlighted in project summary guidance: objectives, results, key activities, and lessons learned should be clearly summarized for stakeholders.

Scope of the Analysis

The notebook simulates 300 shipments across five customers, multiple lanes, four transportation modes, and several origins and destinations. Each shipment includes core transportation cost drivers such as base rate, driver cost, toll cost, fuel surcharge, fixed cost, and accessories.

The analysis covers:

- Cost modeling by shipment.
- Revenue, CM1, and EBIT calculation.
- Win and loss analysis versus competitor price.
- Mode-level profitability.
- Customer-level profitability.
- Loss-making shipment review.
- Price bucket analysis and pricing recommendations.

How the Dataset Was Built

The dataset is simulated rather than sourced from a live transportation management system. That is appropriate for a portfolio project, as long as the assumptions are transparent and the logic is commercially reasonable.

Each shipment was created using:

- A randomly assigned customer, lane, origin, destination, mode, and equipment type.
- A mode-based distance rule, with shorter distances for drayage and store delivery, and longer distances for inland and linehaul.
- Cost logic that reflects transportation economics, including fuel, tolls, driver time, and overhead.
- Margin assumptions that intentionally allow some shipments to be loss-making, especially in drayage and store delivery where pricing pressure is often higher.
- A competitor price and a probability-based win model so the outcome is not unrealistically binary.

Key Calculations

CM1

CM1 is the first profitability layer. It is calculated as revenue minus direct transportation cost. In this project, CM1 shows how much gross profit remains after covering shipment-level cost.

EBIT

EBIT is the second profitability layer. It is calculated after allocating overhead on top of direct transportation cost. This helps show whether a shipment or customer relationship is attractive after operating costs are considered.

Price Difference

Price difference compares the project price to the competitor price. This is used to evaluate whether being cheaper, near parity, or more expensive changes the chance of winning.

Why the Win Model Matters

The project originally faced a common modeling issue: a hard cutoff can create unrealistic outcomes such as 100 percent win rate in some buckets and 0 percent in others. A more realistic project summary should show assumptions, issues, and lessons learned rather than hiding them.

To address that, the final notebook uses a probabilistic win model. When the shipment price is much cheaper than the competitor, the win probability is highest. When it is slightly higher or much higher, the win probability falls sharply. This creates a more realistic pricing story and is much easier to explain in an interview.

Main Findings

1. Price position is the strongest driver of win rate

The price bucket analysis shows a clear commercial pattern:

Price bucket	Meaning	Win rate	Interpretation
Much cheaper	More than 5% below competitor	92.0%	High win rate, but risk of leaving margin on the table
Slightly cheaper	0% to 5% below competitor	85.1%	Strong balance of competitiveness and profitability
Near competitor	0% to 3% above competitor	67.6%	Outcomes depend on service, capacity, and relationship
Slightly higher	3% to 8% above competitor	25.0%	Most bids are lost at this level
Much higher	More than 8% above competitor	17.3%	Very low win rate, likely unsustainable

These results show that the best pricing zone is usually slightly cheaper to near competitor.

This is where the team can still win meaningful business without always being the cheapest option.

2. Inland and linehaul are the most profitable modes

The mode-level analysis shows inland and linehaul have much higher average CM1 and EBIT than drayage and store delivery. Their longer distances create more revenue opportunity while spreading cost more efficiently.

Mode	Shipments	Avg cost	Avg revenue	Avg CM1	Avg EBIT	Win rate
Drayage	83	348.48	386.29	37.80	9.34	0.63
Inland	78	2718.33	3145.92	427.59	182.21	0.59
Linehaul	59	2447.49	2890.24	442.75	223.71	0.66
Store delivery	80	541.01	606.86	65.85	20.68	0.65

This suggests that pricing discipline matters most in drayage and store delivery because those modes are margin-thin and more exposed to aggressive pricing pressure.

3. Customer profitability is not the same as customer revenue

Customer-level analysis shows that higher revenue does not automatically mean stronger economics. Some customers generate large revenue but weaker margin percentage or lower win rate, which is exactly why customer profitability ownership matters.

Customer	Shipments	Total revenue	Total CM1	Total EBIT	Avg CM1 %	Win rate
CUST1	60	95,291.28	13,797.44	6,398.65	0.11	0.65
CUST2	54	91,142.40	10,600.38	3,491.73	0.09	0.70
CUST3	61	96,321.89	13,306.13	6,168.00	0.13	0.70
CUST4	67	121,637.46	16,691.13	7,221.57	0.13	0.54
CUST5	58	92,123.78	13,484.52	6,560.86	0.12	0.57

This is important because a pricing analyst is expected to manage both growth and profitability, not just shipment count or top-line revenue.

4. Loss-making shipments are present and commercially useful to review

The project intentionally includes negative EBIT shipments. That is useful because it creates a realistic review set for repricing decisions, strategic exceptions, and lane-level investigation.

Examples of loss-making shipments in the notebook show that losses appear in inland, drayage, and store delivery. That indicates pricing issues may come from both competitive discounting and cost build problems, not just one specific mode.

What This Project Shows About the Candidate

From a team lead perspective, this project demonstrates several strengths:

- Ability to structure a pricing problem clearly.
- Understanding of transportation cost components.
- Correct use of CM1 and EBIT as separate layers of profitability.
- Ability to translate analysis into pricing recommendations.
- Good communication through simple markdown explanation.
- Awareness that win/loss outcomes are influenced by both price and service factors.

It also shows an important business instinct: not every win is a good win, and not every lost bid is necessarily a pricing failure.

Gaps and Next Improvements

The project is strong for an interview case study, but it can be improved further with a few additions:

Lane-level analysis

A next step would be to analyze profitability and win rate by origin-destination lane. That would bring the project closer to how an actual product pricing team works during tenders and pricing reviews.

Time series analysis

Adding a shipment date would allow trend reporting for win rate, margin movement, and fuel impact over time.

Contract and accessorial detail

The current model includes accessorial cost, but a more advanced version could also simulate detention rules, demurrage, service-level commitments, and legal contract terms that affect pricing.

Tender scenario view

A tender dashboard or scenario table could show how a 2 percent price reduction changes expected win rate and expected EBIT.

Suggested Pricing Actions

Based on this project, the pricing team should consider these actions:

- Reprice shipments in the slightly higher bucket because that is where win rate drops sharply.
 - Review much cheaper wins to check whether prices can be increased without losing the customer.
 - Monitor drayage and store delivery more closely because they are the most margin-sensitive modes.
 - Use customer profitability reviews, not just revenue reviews, when making account decisions.
 - Add lane-level benchmarking to improve tender strategy and follow-up.
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